Final Project Analysis With JMP

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EDA (Deleted the 3 missing data)

Explore distributions of all variables

Identify extreme right-skew in charges

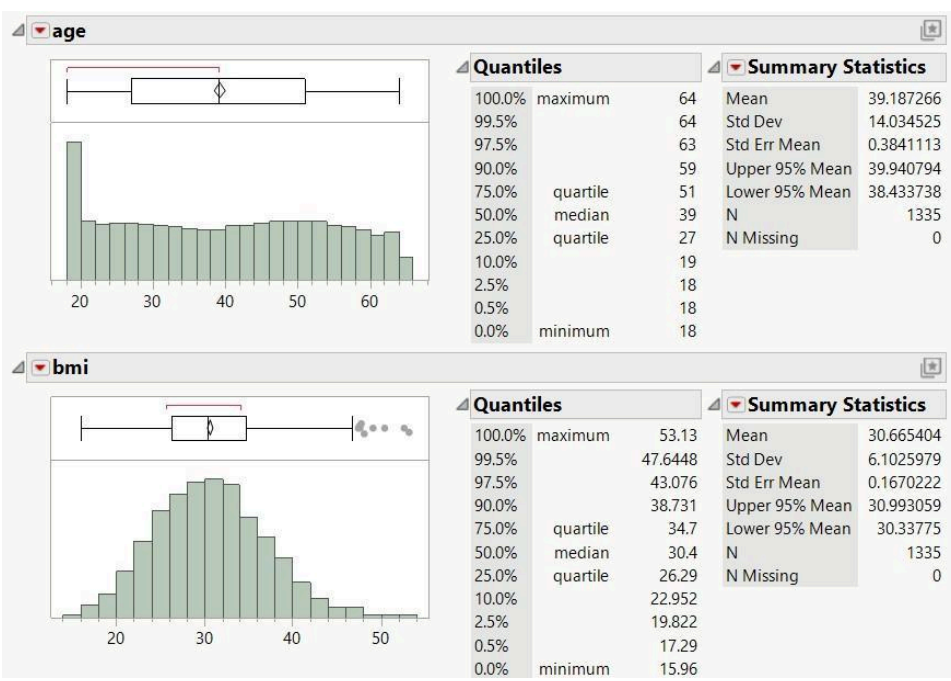
Inspect categorical patterns (sex, region, smoker)

Check scatterplots → BMI × smoker separation

Confirm no multicollinearity among numeric variables

Evidence suggests interaction & log-transform likely

1. **Exploring the quantitative data (age, bmi, children, charges)**



1. Age

Range: 18 to 64

Median: 39

Mean: 39.19

Shape: Slight right-skew

Outliers: None

Interpretation:

The age distribution is fairly uniform across adulthood, with no extreme outliers. This suggests enough variation across ages to detect age effects in the regression model.

2. BMI

Range: 15.96 to 53.13

Median: 30.4

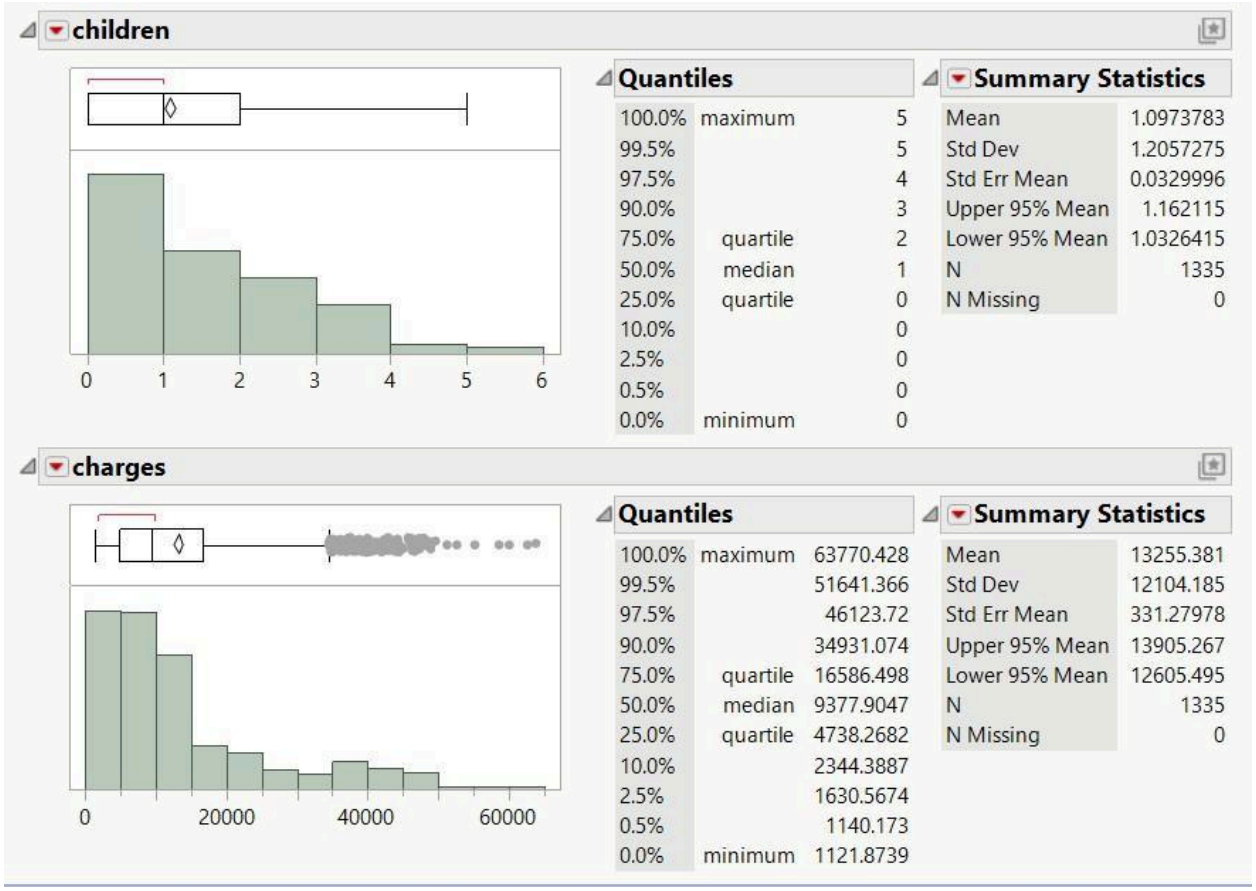
Mean: 30.67

Shape: Slight right-skew

Outliers: **Several BMI values above ~45 appear as legitimate high-BMI individuals**

Interpretation:

Most individuals fall around BMI 25–35 (overweight to obese), with a few naturally high-BMI cases. These are not data errors and should not be removed. Because BMI spreads well across the sample, it is expected to be an important predictor of charges.



Quantiles

100.0%	maximum	5
99.5%		5
97.5%		4
90.0%		3
75.0%	quartile	2
50.0%	median	1
25.0%	quartile	0
10.0%		0
2.5%		0
0.5%		0
0.0%	minimum	0

Summary Statistics

Mean	1.0973783
Std Dev	1.2057275
Std Err Mean	0.0329996
Upper 95% Mean	1.162115
Lower 95% Mean	1.0326415
N	1335
N Missing	0

Quantiles

100.0%	maximum	63770.428
99.5%		51641.366
97.5%		46123.72
90.0%		34931.074
75.0%	quartile	16586.498
50.0%	median	9377.9047
25.0%	quartile	4738.2682
10.0%		2344.3887
2.5%		1630.5674
0.5%		1140.173
0.0%	minimum	1121.8739

Summary Statistics

Mean	13255.381
Std Dev	12104.185
Std Err Mean	331.27978
Upper 95% Mean	13905.267
Lower 95% Mean	12605.495
N	1335
N Missing	0

3. Children

Range: 0 to 5

Median: 1

Mean: 1.10

Shape: Highly right-skewed / discret

Most common values:

0 children → very common

1–2 children → moderate

3–5 children → rare

Interpretation:

The children variable is heavily skewed toward fewer dependents. This variable should still be included, but its effect is likely smaller than age, BMI, or smoker. The lack of symmetry is not a problem because children is a predictor, not a response.

4. Charges

Range: 1,121.87 to 63,770.43

Median: 9,377.90

Mean: 13,255.38

Shape: **Strong right-skew (very heavy tail)**

Outliers: Many high-cost observations (especially > $30,000)

Interpretation:

Charges are very right skewed with a long tail. This is typical because a small subset of people incur extremely high medical expenses—especially smokers and individuals with high BMI.

This strong skew supports:

Using residual diagnostics; **Considering log(charges) transformation** if the residuals show heteroscedasticity or non-normality (very likely)

The quantitative variables show a mixture of symmetric and right-skewed distributions. Age and BMI are moderately right-skewed but overall well-behaved. The 'children' variable is discrete and heavily skewed toward lower values, but this is expected for count data. The most notable feature in the dataset is that the response variable, charges, is strongly right-skewed with many high-cost outliers. These values are legitimate medical expenses and should not be removed. **The strong skew suggests that a log transformation may be needed to meet linear model assumptions.**

Check for the multicollinearity (for the quantitative data)

Correlations

	age	bmi	children	charges
age	1.0000	0.1113	0.0439	0.2964
bmi	0.1113	1.0000	0.0125	0.2002
children	0.0439	0.0125	1.0000	0.0693
charges	0.2964	0.2002	0.0693	1.0000

The correlations are estimated by Row-wise method.

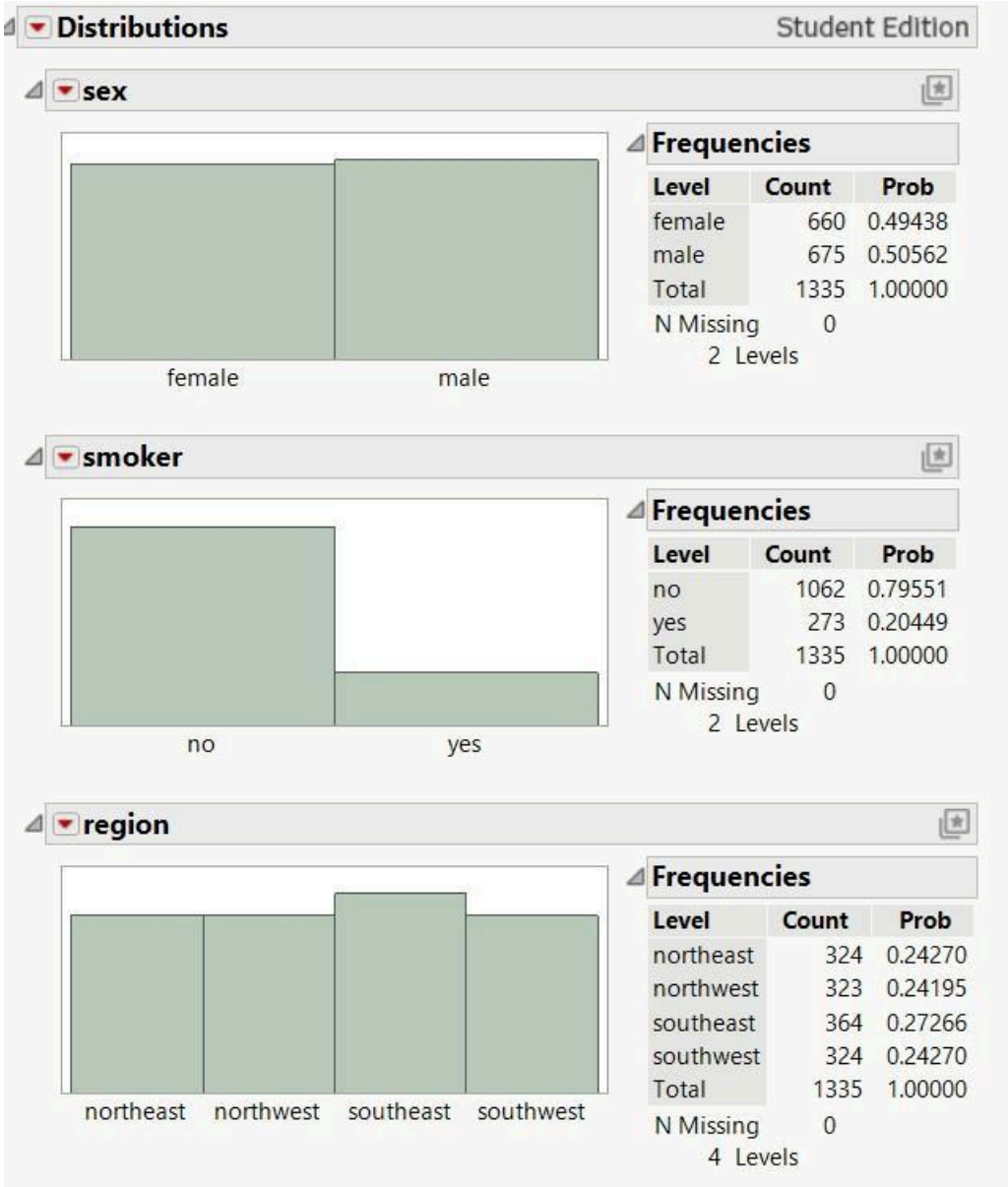
age vs bmi ≈ 0.11 bmi vs children ≈ 0.01 age vs children ≈ 0.04

No correlated predictors → no multicollinearity risk.

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob>|t|	Lower 95%	Upper 95%	VIF
Intercept	-213.5064	942.4549	-0.23	0.8208	-2062.366	1635.3537	.
age	256.85503	11.89585	21.59	<.0001*	233.51836	280.1917	1.0152045
bmi	323.74485	27.32314	11.85	<.0001*	270.14371	377.346	1.0126466
children	477.46406	137.5682	3.47	0.0005*	207.58971	747.33841	1.0020787
smoker[no]	-11925.8	205.4252	-58.05	<.0001*	-12328.8	-11522.81	1.0008868

2.Exploring the qualitative data(sex, smoker, region)



Level	Count	Prob
female	660	0.49438
male	675	0.50562
Total	1335	1.00000

N Missing 0
2 Levels

Level	Count	Prob
no	1062	0.79551
yes	273	0.20449
Total	1335	1.00000

N Missing 0
2 Levels

Level	Count	Prob
northeast	324	0.24270
northwest	323	0.24195
southeast	364	0.27266
southwest	324	0.24270
Total	1335	1.00000

N Missing 0
4 Levels

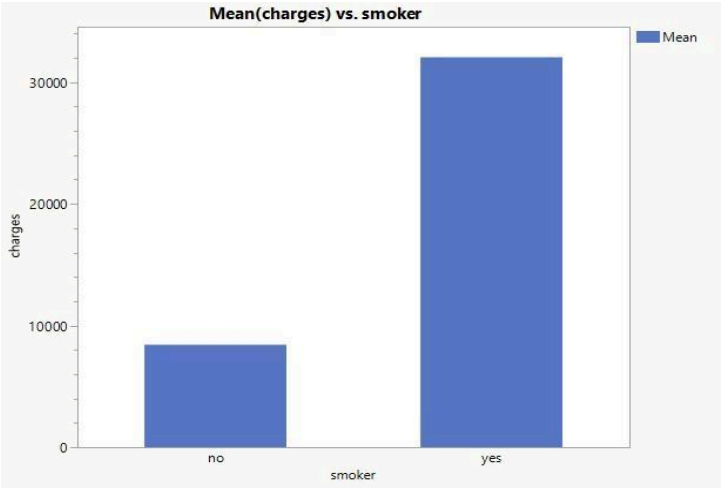
The distributions show that frequency counts, proportions, and bar charts.

- the dataset is balanced (sex, region)
- smoker variable is imbalanced (only 20% smokers)
- some categories are more common (SE region)

These patterns matter for **regression interpretation and interaction effects.**

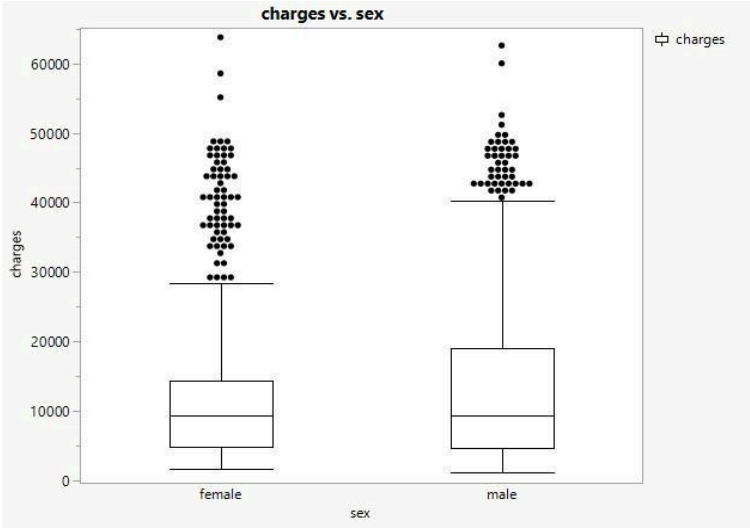
3.Explore relationships

Bar charts : charges vs smoker



The graph shows that the smoker's charges are much higher**, it indicates that interaction BMI * smoker is necessary , smoker is a key predictor**

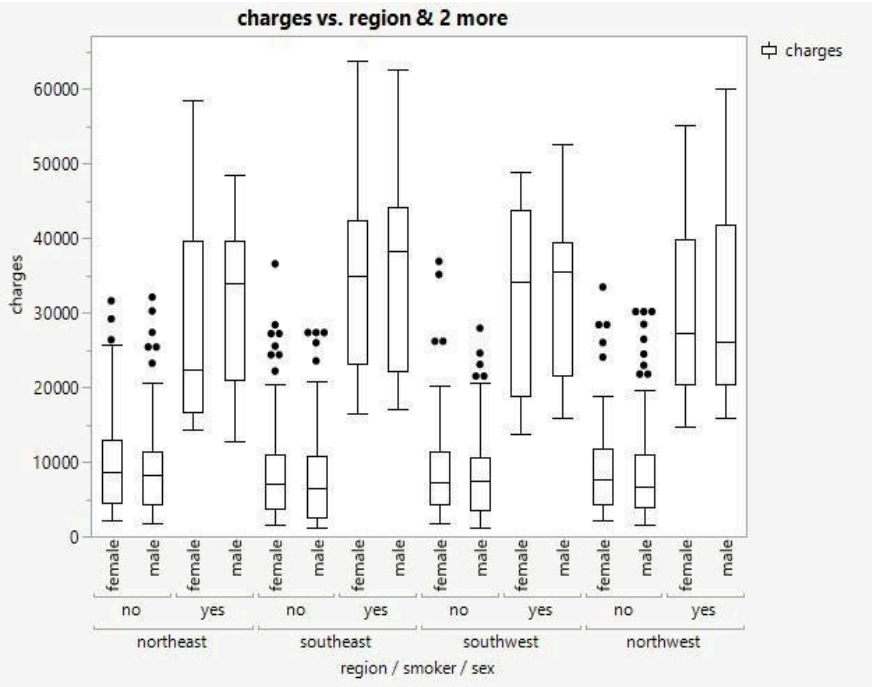
Boxplot : charges vs sex



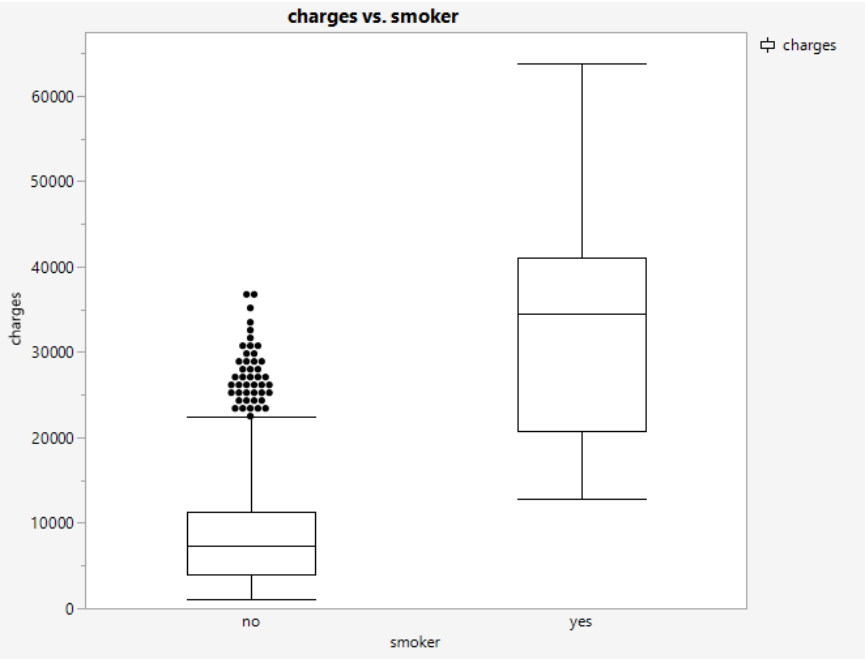
The distributions of charges for males and females are very similar. The medians are nearly the same, and the interquartile ranges (IQRs) overlap substantially, indicating comparable variability in typical charges for both groups. While the male group shows slightly larger spread, these differences are small relative to the overall variability in charges. Both groups exhibit many

high-cost outliers extending to similar levels. Overall**, there is no strong visual evidence that sex has a substantial effect on health insurance charges.**

Boxplots : Charges vs smoker, sex, region



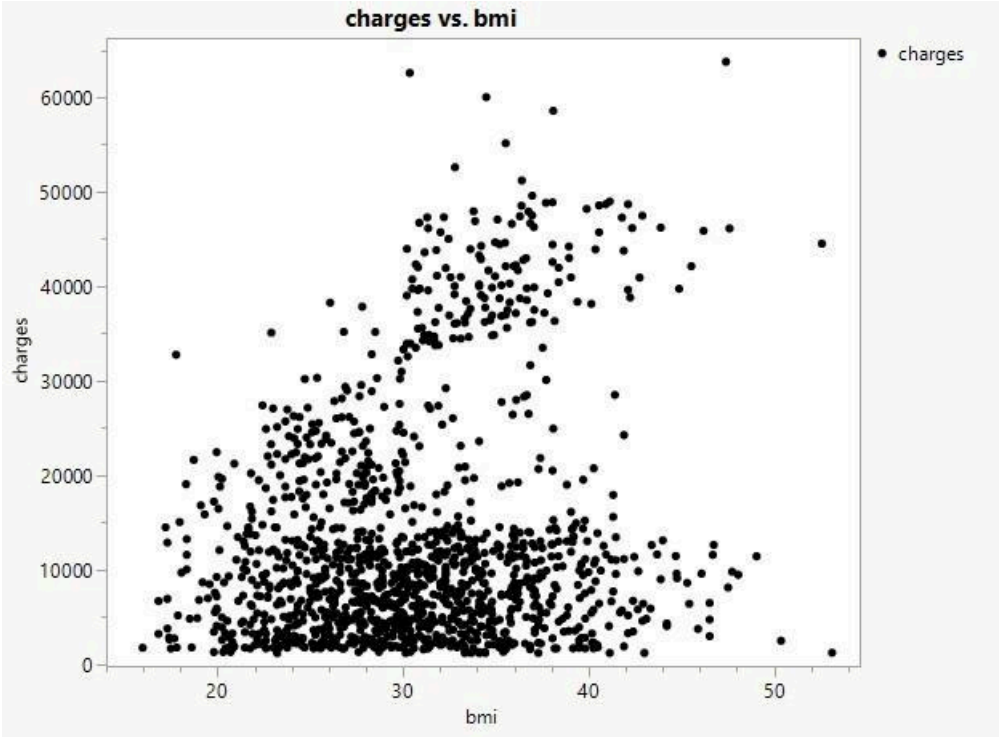
Boxplot smoker vs charges



It shows that smokers always have much higher charges. Smoking is by far the strongest categorical driver of medical charges. **Region does not show major differences in charges. Sex is not a strong predictor of charges.**

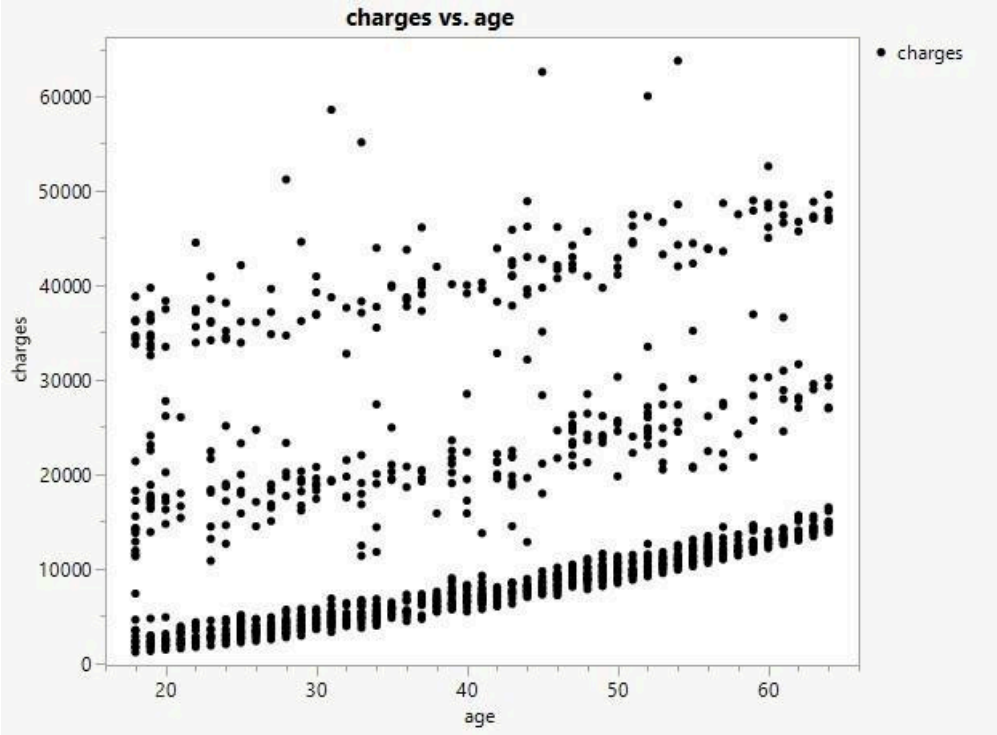
The combined boxplot comparing charges across region, smoker status, and sex reveals clear and consistent patterns. Across all four regions, smokers have substantially higher medical charges than non-smokers, with much higher medians, wider interquartile ranges, and more extreme outliers. This confirms smoking as the strongest categorical predictor in the dataset. In contrast, the differences among regions are relatively small, with similar medians and spreads across Northeast, Northwest, Southeast, and Southwest. Male and female distributions also overlap heavily within each region and smoking group, suggesting that sex has little influence on charges. Overall, smoking dominates the variation in medical costs, while region and sex show minor effects.

4.Check Scatterplots for potential interactions

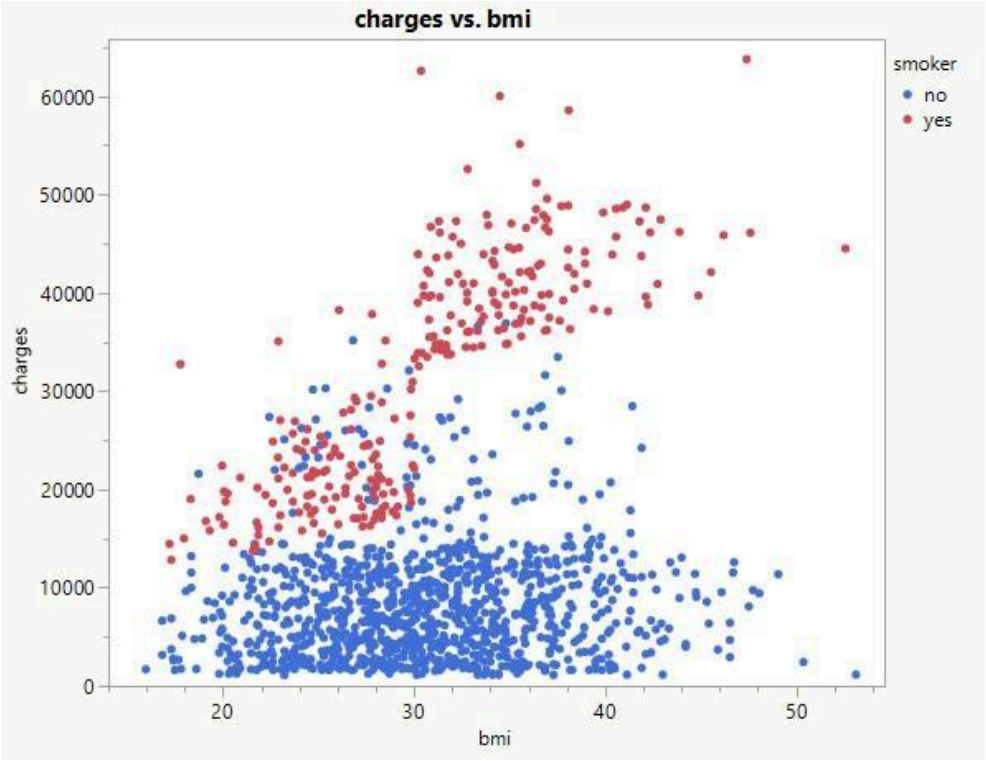


Charges vs BMI

Scatterplot charges vs age



Charges vs BMI colored by smoker



EDA FINDINGS

Numerical Summary

Age, BMI, and charges show typical skew patterns. Charges are strongly right-skewed with a long upper tail, suggesting that a log transformation may improve normality and stabilize variance.

Categorical Comparisons

Smokers display dramatically higher charges compared to non-smokers, indicating a strong main effect. Regions show moderate differences, and sex shows only small variation.

Scatterplots

Charges increase with age and BMI, but the rate of increase appears much steeper among smokers. This visual pattern supports including a **BMI × smoker** interaction in the regression model.

Correlation

Age and BMI are weakly correlated, reducing multicollinearity risks. Charges correlate moderately with BMI and strongly with smoker status.

Initial Findings

Charges skewed → may need log transform

Smokers have much higher charges

Visual interaction between BMI and smoker

No strong multicollinearity

VARIABLE SCREENING

Run Stepwise

Current Estimates

Lock	Entered	Parameter	Estimate	nDF	SS	"F Ratio"	"Prob>F"
☑	☑	Intercept	-213.50638	1	0	0.000	1
☐	☑	age	256.855032	1	1.71e+10	466.214	7e-89
☐	☐	sex{female-male}	0	1	4294573	0.117	0.73217
☐	☑	bmi	323.744854	1	5.142e+9	140.393	7.3e-31
☐	☑	children	477.46406	1	4.412e+8	12.046	0.00054
☐	☑	smoker{no-yes}	-11925.805	1	1.23e+11	3370.294	0
☐	☐	region{northwest&southwest&northeast-southeast}	0	1	82679699	2.260	0.13303
☐	☐	region{northwest&southwest-northeast}	0	1	94989373	2.597	0.10733
☐	☐	region{northwest-southwest}	0	1	52589692	1.436	0.23095

Step History

Step	Parameter	Action	"Sig Prob"	Seq SS	RSquare	Cp	p	AICc	BIC	
1	smoker{no-yes}	Entered	0.0000	1.21e+11	0.6210	695.36	2	27600	27615.5	○
2	age	Entered	0.0000	1.98e+10	0.7221	156.77	3	27187.7	27208.5	○
3	bmi	Entered	0.0000	5.165e+9	0.7485	17.491	4	27056.4	27082.3	○
4	children	Entered	0.0005	4.412e+8	0.7508	7.4233	5	27046.3	27077.5	◉

These **4 predictors** survived Stepwise screening → meaning they were statistically useful for predicting **charges**.

Sex and region dummy variables (3 region categories)

Stepwise tested them but rejected them—they did **not** improve AIC/BIC/p-values enough to stay in the model.

All Possible Models

	Model	Number	RSquare	RMSE	AICc	BIC	Cp
1	smoker{no-yes}	1	0.6210	7454.9003	27599.9761	27615.5481	695.3565
2	age	1	0.0878	11564.7108	28772.3376	28787.9097	3545.4271
3	bmi	1	0.0401	11863.5935	28840.4655	28856.0375	3800.7406
4	age,smoker{no-yes}	2	0.7221	6385.9151	27187.7323	27208.4890	156.7733
5	bmi,smoker{no-yes}	2	0.6598	7065.6636	27457.8085	27478.5652	489.9140
6	children,smoker{no-yes}	2	0.6249	7418.8873	27588.0568	27608.8135	676.3206
7	age,bmi,smoker{no-yes}	3	0.7485	6076.9961	27056.3547	27082.2930	17.4913
8	age,children,smoker{no-yes}	3	0.7245	6360.9491	27178.2858	27204.2241	146.0719
9	age,smoker{no-yes},region{northwest&southwest&northeast-southeast}	3	0.7226	6382.0421	27187.1249	27213.0632	155.8575
10	age,bmi,children,smoker{no-yes}	4	0.7508	6051.9353	27046.3359	27077.4528	7.4233
11	age,bmi,smoker{no-yes},region{northwest&southwest&northeast-southeast}	4	0.7490	6073.4926	27055.8297	27086.9465	16.9325
12	age,bmi,smoker{no-yes},region{northwest-southwest}	4	0.7488	6076.1672	27057.0052	27088.1220	18.1147
13	age,bmi,children,smoker{no-yes},region{northwest&southwest&northeast-southeast}	5	0.7512	6049.0717	27046.0892	27082.3816	7.1618
14	age,bmi,children,smoker{no-yes},region{northwest&southwest-northeast}	5	0.7511	6050.3314	27046.6452	27082.9376	7.7159
15	age,bmi,children,smoker{no-yes},region{northwest-southwest}	5	0.7510	6051.1547	27047.0085	27083.3009	8.0780
16	age,bmi,children,smoker{no-yes},region{northwest&southwest&northeast-southeast},region{northwest...	6	0.7517	6045.4358	27045.5032	27086.9681	6.5636
17	age,bmi,children,smoker{no-yes},region{northwest&southwest&northeast-southeast},region{northwest-...	6	0.7515	6048.1014	27046.6802	27088.1451	7.7346
18	age,bmi,children,smoker{no-yes},region{northwest&southwest-northeast},region{northwest-southwest}	6	0.7513	6049.5582	27047.3233	27088.7882	8.3748
19	age,bmi,children,smoker{no-yes},region{northwest&southwest&northeast-southeast},region{northwest...	7	0.7519	6044.4358	27046.0831	27092.7174	7.1251
20	age,sex{female-male},bmi,children,smoker{no-yes},region{northwest&southwest&northeast-southeast},r...	7	0.7517	6047.4332	27047.4068	27094.0411	8.4406
21	age,sex{female-male},bmi,children,smoker{no-yes},region{northwest&southwest&northeast-southeast},r...	7	0.7515	6050.1046	27048.5860	27095.2203	9.6137
22	age,sex{female-male},bmi,children,smoker{no-yes},region{northwest&southwest&northeast-southeast},r...	8	0.7520	6046.4294	27047.9875	27099.7882	9.0000

As the carefully compared all the critical numbers , we also choose for the Model 10 as these 4 variables as the best model.

Because it has **very high R^2** and **excellent Cp; the lowest or nearly lowest AICc** ; **lowest complexity;** Region and sex add complexity without improving fit. This is the "simplest model nearly as good as full model"

Stepwise Screening Conclusion

Stepwise regression (mixed direction, AICc stopping rule) selected **age, BMI, children, and smoking status as important predictors of insurance charges.** The variables sex and region were evaluated but not retained due to non-significant contribution to the model. Smoking status had the largest effect, increasing average charges by approximately $12,000, followed by BMI and age. The selected subset forms a strong and interpretable starting model for the final regression analysis.

CONCLUSION

Exploratory analysis revealed that charges are heavily right-skewed with many high-cost cases, primarily among smokers and people with high BMI. Age, BMI, and number of children showed clear positive associations with charges, while smoker status exhibited the strongest separation between groups. In contrast, sex and region displayed little to no difference in charge distributions. Correlation analysis confirmed weak relationships among predictors and ruled out multicollinearity.

Stepwise regression supported these findings by **selecting age, BMI, children, and smoking as the most important predictors**, while rejecting sex and region due to their limited contribution to the model. Smoking had the largest estimated effect, followed by BMI and age. The resulting subset aligns with the EDA and provides a well-supported foundation for Model 1 .

Model 1:

This model includes only the main effects

$E(y) = \beta 0 + \beta 1x1 + \beta 2\ x2 + \beta 3x3 + \beta 4x4$

Where x_1= age

x_2 = bmi

x_3 = children

x_4 =smoker[no]

Smoker Type (2 levels: yes(baseline), no)

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob>|t|	Lower 95%	Upper 95%	VIF
Intercept	-213.5064	942.4549	-0.23	0.8208	-2062.366	1635.3537	.
age	256.85503	11.89585	21.59	<.0001*	233.51836	280.1917	1.0152045
bmi	323.74485	27.32314	11.85	<.0001*	270.14371	377.346	1.0126466
children	477.46406	137.5682	3.47	0.0005*	207.58971	747.33841	1.0020787
smoker[no]	-11925.8	205.4252	-58.05	<.0001*	-12328.8	-11522.81	1.0008868

$$\mathbf{E(y) = 11712.298 + 256.85503x_1 + 323.74485x_2 + 477.46406x_3 - 23851.61x_4}$$

β1= 256.86, age, p < .0001: For each additional year of age, expected charges increase by about $256.86, holding all other variables constant.

B2 = 323.74, bmi, p < .0001: Each 1-unit increase in BMI is associated with an increase of approximately $323.74 in expected charges.

B3 = 477.46, children, p = 0.0005: Each additional dependent increases expected charges by about $477.46.

β4 = –23851.61,Smoker (no vs yes), p < .0001): Non-smokers have $23851.61 lower charges than smokers on average.

This is the **largest** and most significant effect in the model.

Interpretation of Coefficients (Model 1).

Using indicator coding with smokers as the reference group, age, BMI, number of children, and smoking status were all statistically significant predictors of medical charges. Controlling for other variables, each additional year of age increased expected charges by approximately **$257**, while each one-unit increase in BMI increased charges by about **$324**. Each additional child raised annual charges by roughly **$477**. Smoking had by far the largest effect: **non-smokers had approximately $23,852 lower charges than smokers**, indicating that smokers incur dramatically higher medical costs compared to non-smokers.

1. Global F-test (Overall Usefulness)

Conduct the global F-test of model useful at the $\alpha = 0.05$ **level of significance.**

$H0$**:** $\beta 1 = \beta 2 = \cdots = \beta 4 = 0$ $vs\ Ha$**:** $At\ least\ one\ of\ the\ \beta i \neq 0$ i= 1 to4

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	4	1.4673e+11	3.668e+10	1001.569
Error	1330	4.8712e+10	36625921	Prob > F
C. Total	1334	1.9545e+11		<.0001*

F =1001.569 , p_value = 0.0001 < 0.05, reject the $H0$. The data provides strong evidence that at least one of the model coefficients is nonzero. The overall model appears to **be statistically useful** for predicting charges.

2.Inferences About the Individual β Parameters(T-test for individual)

Predictor	Estimate	p-value	Significant?
Intercept	11712.30	<.0001	
Age	256.86	<.0001	Yes
BMI	323.74	<.0001	Yes
Children	477.46	0.0005	Yes
Smoker (no)	–23851.61	<.0001	Yes (strongest)

3. Perform all diagnostics

Model Evaluation

Summary of Fit

RSquare	0.750763
RSquare Adj	0.750013
Root Mean Square Error	6051.935
Mean of Response	13255.38
Observations (or Sum Wgts)	1335

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	4	1.4673e+11	3.668e+10	1001.569
Error	1330	4.8712e+10	36625921	Prob > F
C. Total	1334	1.9545e+11		<.0001*

The estimator of σ^2 is s^2 = 36625921 is the **Mean Square for Error (MSE)**
estimated standard error of the regression model: s = RMSE = 6051.94

We expect the model to provide predictions of charges to within about ±
$2s$ = ±12, 103. 88dollars.

The RMSE of 6046.429 indicates that typical prediction errors are about $6,046. Since prediction error is approximately ±2×RMSE, we expect predictions to vary by roughly ±$12,103.88around the fitted line

Measure of Model Adequacy

The **Adjusted R^2 = 0.7508** , 75.08% of the total variatin in charges y is explained by the model with variables age , sex,bmi, children,etc , adjusting for the sample size and the number of vairables in the Model1.

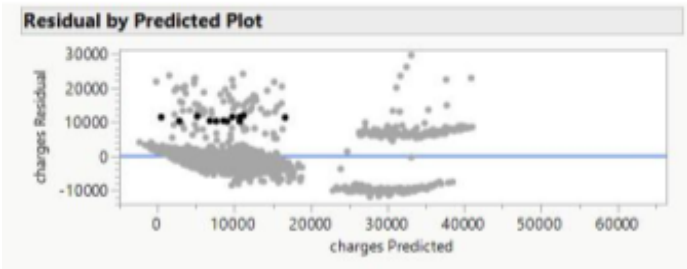
Check Practical Utility using Coefficient of Variation (CV)

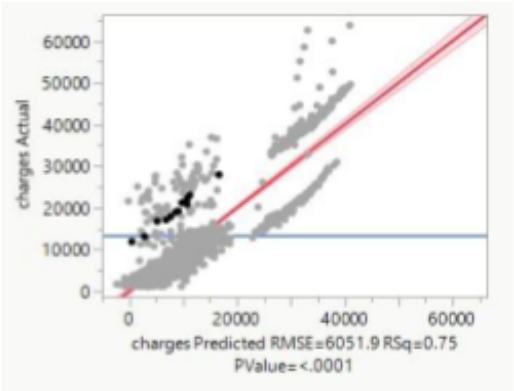
$$CV = 6051.94/13255.38 \times 100 \approx \mathbf{45.66\%}$$

The desired rule of thumb is that regression models should have a CV of 10% or smaller. Since the calculated CV is 45.66%, the value of s is likely too large relative to the mean of the dependent variable. Even though the model is statistically useful (high F-ratio), it is considered undesirable for practical use in predicting the absolute dollar value of charges,. **This finding strongly suggests that a transformation of the dependent variable (y) is necessary to reduce prediction error, confirming the EDA suggestion to consider ln(charges)**

Residual Diagnostics

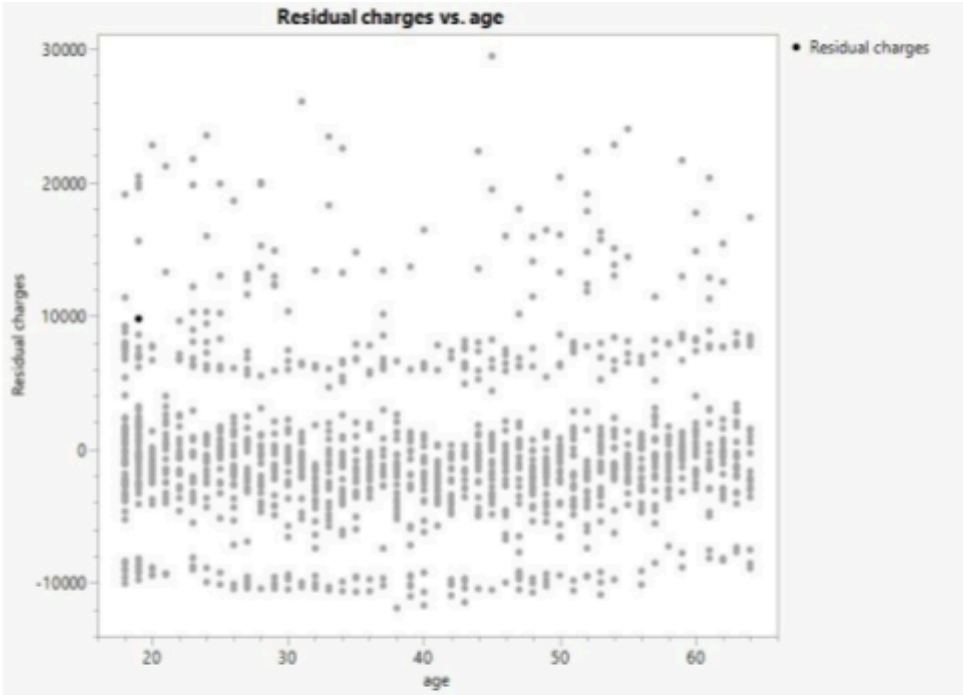
The residual by predicted plot

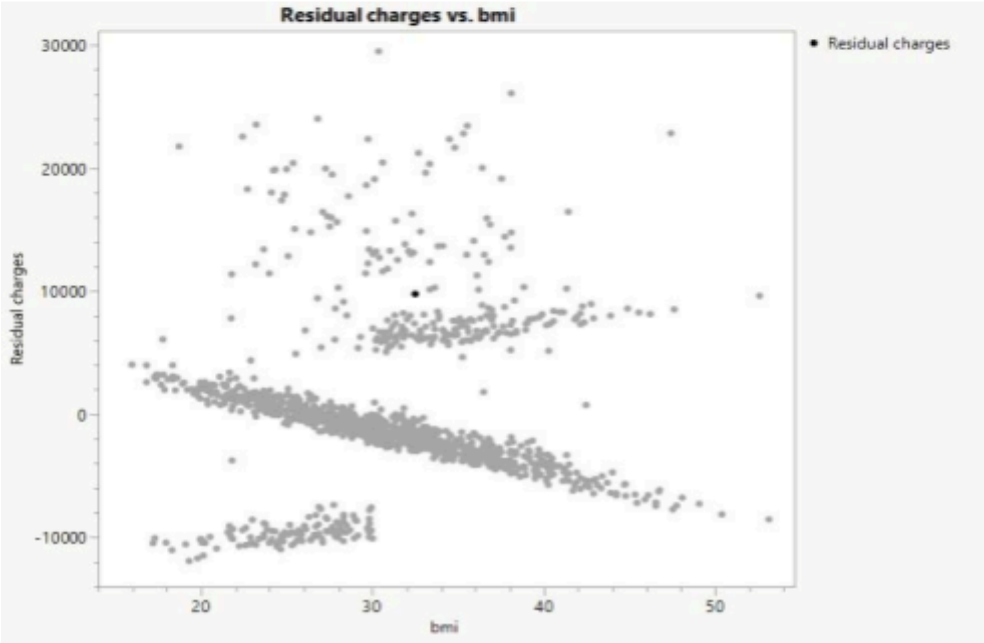


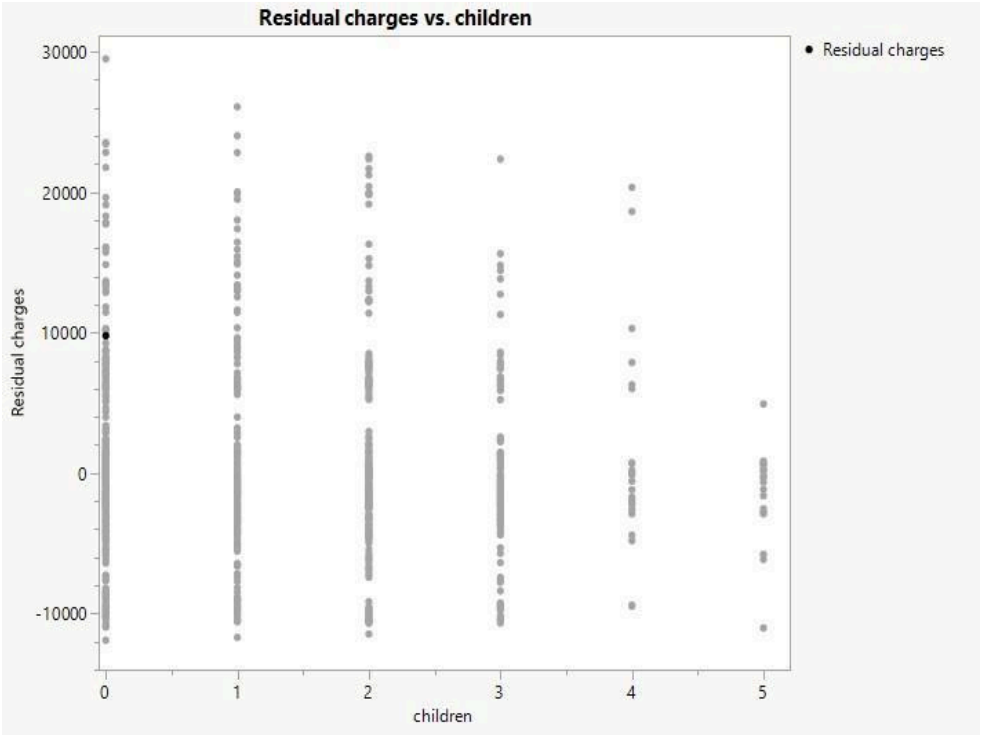


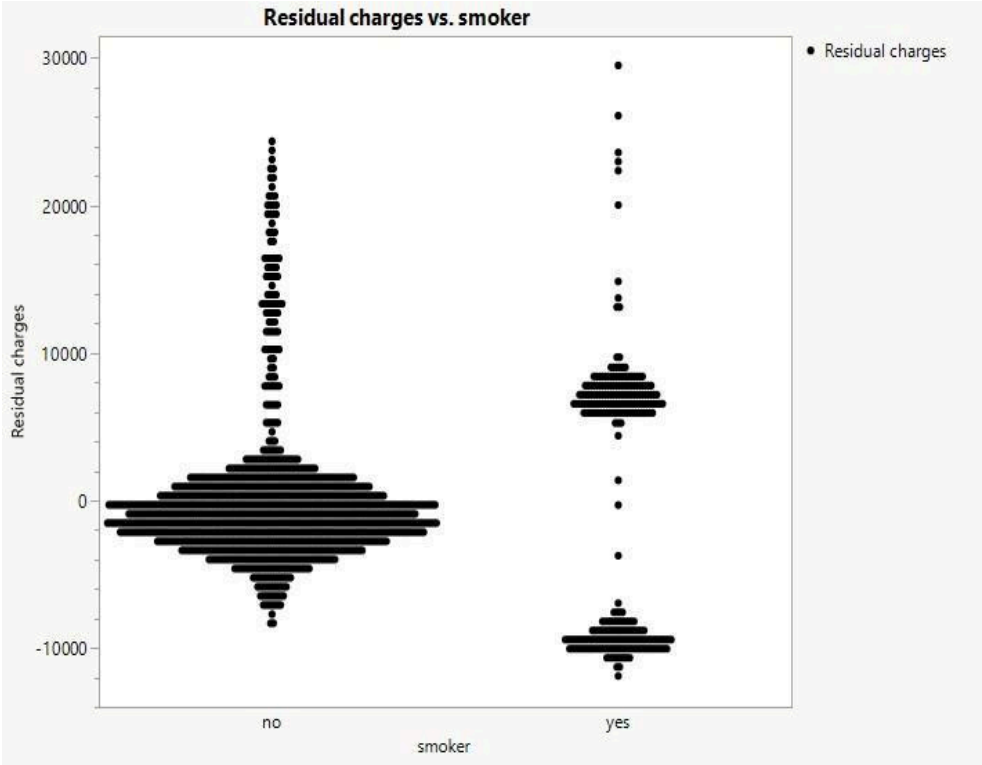
The residual by predicted plot shows **two distinct horizontal bands of residuals separated around predicted** charges of $20,000. Within these bands, the variability of the upper band (high predicted charges) is visibly much greater than the lower band. This pattern, where the size of the residuals varies across the range of ^y(not showing a random, consistent width), **is a strong indicator of heteroscedasticity (unequal variances)**

When the variance increases with the mean response (as suggested by the wider spread at higher predicted charges), it **indicates that a multiplicative model may be appropriate**. This means transforming the response variable, y, often **using the natural logarithm (ln(y)),** is necessary to stabilize the variance .


Residual charges vs. age
Residual charges
30000
20000
10000
0
-10000
Residual charges
20
30
40
50
60
age


Residual charges vs. bmi
Residual charges
30000
20000
10000
0
-10000
Residual charges
20
30
40
50
bmi



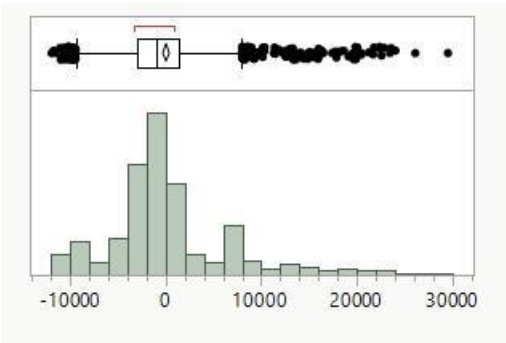


The Residual charges vs. bmi plot **clearly shows two distinct fan-shaped patterns (a large, positive cluster and a smaller, negative cluster)**, rather than a random scatter. This structure indicates that the linear model **is misspecified** and that the effect of BMI depends significantly

on some other variable in the model (almost certainly the smoker variable, which creates two populations). This **strongly suggests the need for an interaction term.**

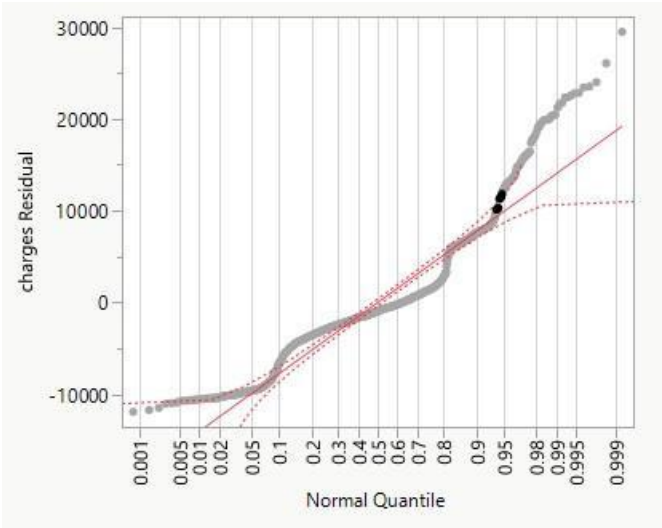
Normality diagnostics

Residual Histogram



It shows that the distribution **is heavy tailed right skewed with outliers** as well , the normality assumption would be violated.

Normal Probability Plot



The QQ plot shows a severe **S-shaped** deviation from the diagonal line, particularly at both tails (residuals dropping sharply negative at the low end and curving sharply positive at the high end). This pattern confirms that the residual distribution **is highly right-skewed and not normal**,

which is a severe violation of the normality assumption A nonliear trend indicates that the normality assumption is most likely violated.

Sharpiro-Wilk Test

$H0$: The probability distribution of ε is normal

Ha: The probability distribution of ε is not normal

Goodness-of-Fit Test

	W	Prob<W
Shapiro-Wilk	0.9005086	<.0001*

	A^2	Simulated p-Value
Anderson-Darling	43.408222	<.0001*

Note: Ho = The data is from the Normal distribution. Small p-values reject Ho.

P_vaule <0.0001 < 0.05 , reject the $H0$. There is significant evidence that probability distribution of ε is not normal.

Residual diagnostics for the updated Model 1 (**including age, BMI, children, and smoker only**) reveal substantial violations of regression assumptions. The residual-by-predicted plot shows a clear **pattern of heteroscedasticity**, with residuals becoming increasingly spread out as predicted charges increase. This indicates that the variance of the errors is not constant, especially for high-cost individuals, a common issue in healthcare cost data.

Additionally, the residuals display distinct curved and clustered patterns that correspond to different smoker and BMI groups. **These patterns show that Model 1 does not adequately capture the different BMI–charge relationships between smokers and non-smokers, indicating model misspecification and the need for an interaction term.** This is further supported by residuals plotted against BMI, where two distinct trends appear, one for smokers and one for non-smokers.

Normality diagnostics (histogram and Q–Q plot) show **strong right-skew and long positive tails,** confirming that the error distribution is not normal. This is expected when modeling highly skewed cost data in raw dollar units.

Because the insurance dataset does not contain replicated predictor combinations, a formal lack-of-fit test cannot be performed; however, the diagnostic plots provide clear evidence of lack of fit. Taken together, these findings indicate that **Model 1 is structurally incomplete and statistically inadequate.**

To correct these issues, two steps are required:

Structural Correction (Model 2):

Add the BMI × smoker interaction term to correctly model the different cost behavior of smokers with increasing BMI. This interaction is strongly suggested by EDA and by the residual patterns in Model 1.

Variance and Normality Correction (Model 3):

Fit a model **using log(charges) as the response to address heteroscedasticity and non-normality**. A log transformation stabilizes variance, reduces skewness, and produces more normally distributed residuals, resulting in a more appropriate linear modeling framework.

After fitting the log-transformed model, residual diagnostics should be repeated to confirm that the corrective actions have resolved the identified problems.

Validation Techniques

Used only to show baseline

Failed assumptions

No validation performed

Model 2 Structural correction (add interaction BMI×smoker)

Does the effect of BMI on charges differ between smokers and non-smokers?

$$E(y) = \beta 0 + \beta 1x1 + \beta 2x2 + \beta 3x3 + \beta 4x4 + \beta 5x2x4$$

Indicator Function Parameterization

Term	Estimate	Std Error	t Ratio	Prob>|t|	Lower 95%	Upper 95%
Intercept	-22847.84	1499.047	-15.24	<.0001*	-25788.6	-19907.09
age	264.00987	9.545404	27.66	<.0001*	245.28417	282.73557
bmi	1438.6853	46.52146	30.93	<.0001*	1347.4218	1529.9488
children	513.60664	110.3528	4.65	<.0001*	297.12202	730.09126
smoker[no]	20065.293	1649.626	12.16	<.0001*	16829.137	23301.448
bmi*smoker[no]	-1430.67	52.65624	-27.17	<.0001*	-1533.969	-1327.372

$E(y) = -22847.84+264.00987x1+1438.6853x_2+513.60664x_3+20065.293x_4 – 1430.67x_2x_4$

Evaluating Model 2 Utility and Structure

Model Evaluation and Practical Utility

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	5	1.6413e+11	3.283e+10	1393.028
Error	1329	3.1317e+10	23564363	Prob > F
C. Total	1334	1.9545e+11		<.0001*

Summary of Fit

RSquare	0.839766
RSquare Adj	0.839164
Root Mean Square Error	4854.314
Mean of Response	13255.38
Observations (or Sum Wgts)	1335

The estimator of σ^2 is s^2 = 23564363 is the **Mean Square for Error (MSE) estimated standard error of the regression model**: s = RMSE = 4854.314

We expect the model to provide predictions of charges to within about ± $2s$ = ±9,708.628dollars. s is the estimated standard error of the regression model. We expect most (approximately 95%) of the observed y-values to lie within 2s of their predicted values ($\hat{y}$), We expect predictions of charges to vary by roughly ±**$9,708.628**around the fitted line.

Measure of Model Adequacy

The Adjusted R^2 =0.8392, meaning 83.92% of the total variation in charges (y) is explained by Model 2 (incorporating the interaction term), adjusting for the sample size and the number of variables in the model". The improvement from 75.08% to 83.92% demonstrates that the interaction term significantly improved model adequacy.

Check Practical Utility using Coefficient of Variation (CV)

$$CV = 4854.31/13{,}255.38 \times 100 \approx \mathbf{36.62\%}$$

Since the CV remains far above the desirable 10% or smaller rule of thumb, the model's prediction error (s) is still too large relative to the mean of the response variable. This high CV, combined with the severe assumption violations (heteroscedasticity and non-normality) documented in Model 1 and which are not solved simply by adding an interaction term, **strongly validates the need to proceed to the next step: transformation of the dependent variable (Model 3)**

1. Test the overall utility of the model using the global F-test at α= 0.05

The global F -test is used to test the null hypothesis

H**0:** β**1** =β**2** =… β**5 = 0** vs

Ha**:** $at\ least\ one\ of\ the\ \beta j \neq 0$ **Where** j **= 1-5**

Conclusion: The test statistic F = 1393.028 and p-value<0.0001< 0.05, there is sufficient evidence to conclude that the model fit is **a statistically useful** predictor of charges.

2. **Testing the Interaction Term (Specific Utility)**

To evaluate whether the effect of BMI on charges differs between smokers and non-smokers, we tested the BMI × smoker interaction coefficient, β5

Hypotheses:

H0: β5=0 (no interaction, effect of BMI is independent of smoker status) versus

Ha: β5≠0 (interaction exists)

In Model 2, the estimated interaction coefficient is β5=−1430.67with p < 0.0001, indicating a highly significant interaction. This shows that the effect of BMI on charges depends strongly on smoking status, and the interaction term must be retained in the model.

x4=smoker[no] so smokers (yes) have x4=0 , non-smokers (no) have x4=1.

Grouping BMI terms, the fitted model can be written as

E(y)=(β0+β1x1+β3x3+β4x4)+**(β2+β5x4)x2.**

For smokers (**baseline, x4=0)**, the estimated BMI slope is β2=1438.69, so each 1-unit increase in BMI increases charges by about $1,439. For non-smokers (x4=1), the BMI slope is β2+β5=1438.69−1430.67≈8.02, so BMI has almost no effect on charges. This large difference in slopes confirms that BMI strongly affects charges among smokers but not among non-smokers, consistent with the significant interaction and the partial F-test comparing Models 1 and 2.

Partial F_Test

"Does adding the BMI × smoker interaction significantly improve the model's ability to predict charges?"

H_0: all added parameters = 0 → here, $\beta_5 = 0$ (no interaction)

H_a: at least one added parameter ≠ 0 → here, $\beta_5 \neq 0$ (interaction needed)

F-ratio = 738.21, p-value = 1.1816e-129; we reject H0, indicating **highly significant evidence** that the interaction terms contribute to the prediction.

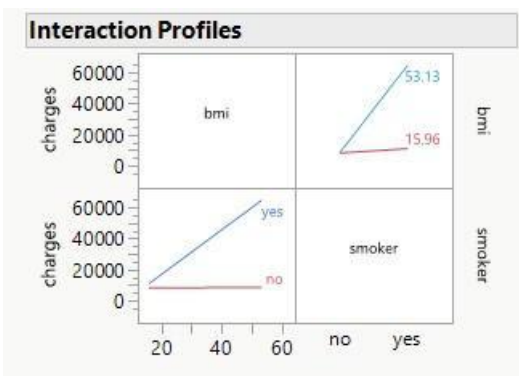
Custom Test

Parameter	
Intercept	0
age	0
bmi	0
children	0
smoker[no]	0
bmi*smoker[no]	1
=	0

Value	-715.3352394
Std Error	26.328117836
t Ratio	-27.17001055
Prob>|t|	1.1816e-129
SS	17395436113

Sum of Squares	17395436113
Numerator DF	1
F Ratio	738.20947332
Prob > F	1.1816e-129

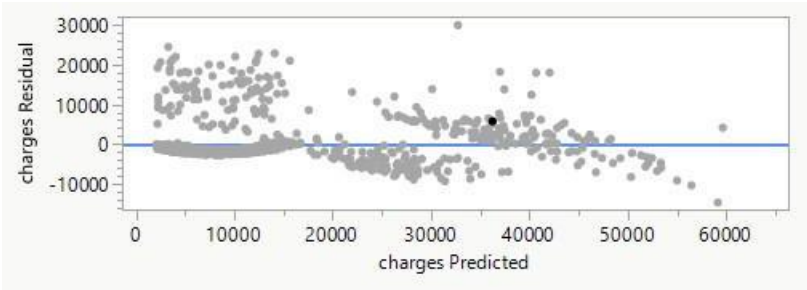
Interaction Plot



The interaction plot shows non-parallel lines for smokers and non-smokers across BMI. Smokers exhibit a much steeper positive relationship between BMI and charges, while the slope for non-smokers is nearly flat. This visually confirms the strong BMI × smoker interaction detected in the regression model.

Re-Evaluation of Model Assumptions (Residual Analysis)

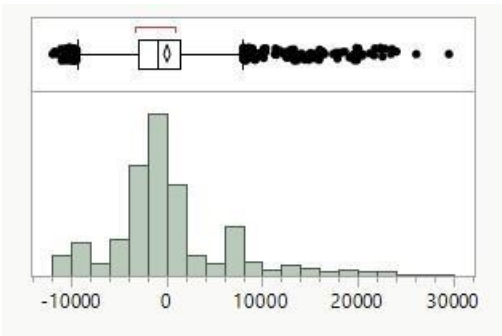
Detecting Lack of fit



The residuals vs predictor plot shows that the residuals to now show a **random scatter** around zero, confirming Model 2 is now is bettet than Model 1.

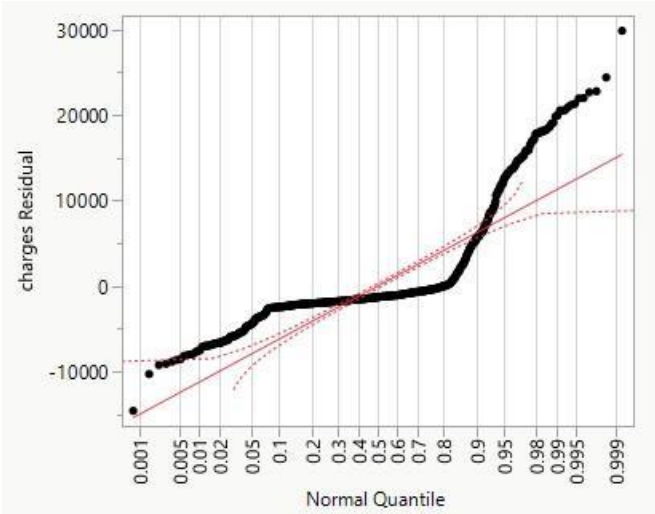
Normality Check

Histogram of residuals



It shows that the distribution is right skewed with outliers as well , the normality assumption would be violated.

Normal Quantile Plot



The QQ plot shows that there is a curvilinear, it indicates that the model still violated the normality assumptions.

Sharpiro-Wilk Test

H_0: The probability distribution of ε is normal

H_a: The probability distribution of ε is not normal

Goodness-of-Fit Test

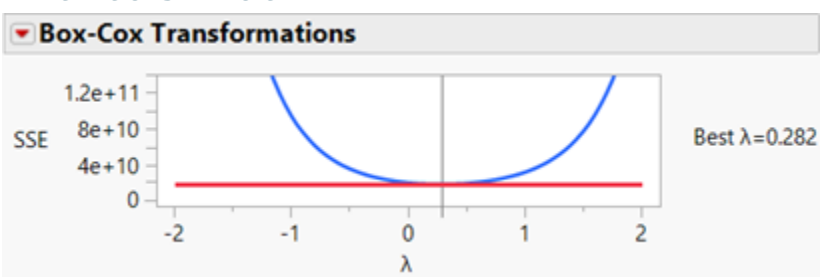
	W	Prob<W
Shapiro-Wilk	0.9005086	<.0001*

	A^2	Simulated p-Value
Anderson-Darling	43.408222	<.0001*

Note: Ho = The data is from the Normal distribution. Small p-values reject Ho.

P_vaule <0.0001 < 0.05 , reject the H_0. There is significant evidence that probability distribution of ε is not normal.

Box-tox transformaiton

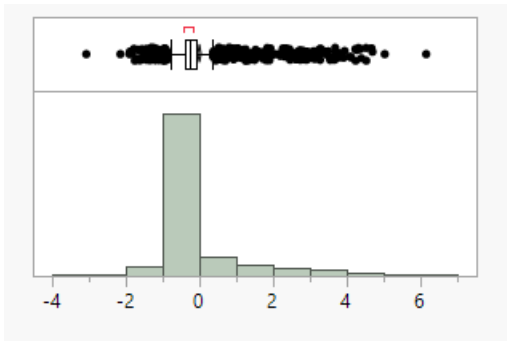


minimum SSE.λ = 0 corresponds to log transformation. (λ =0.282)

The SSE curve (blue) reaches its minimum at λ = 0. This indicates that we are going to use a natural log transformed response mode.

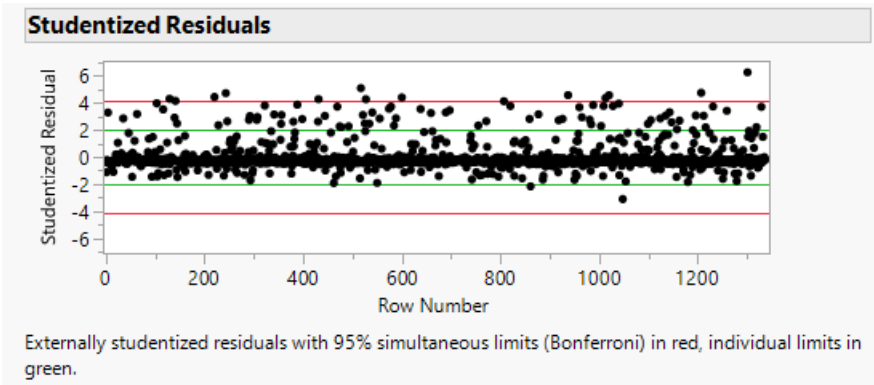
Outlier and Influence observation

Using studentized residual



Quantiles

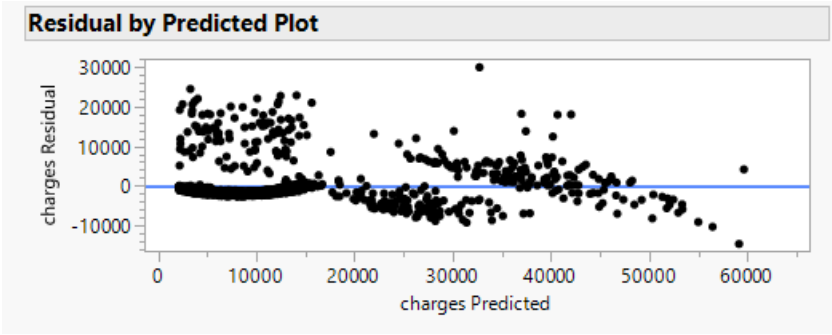
100.0%	maximum	6.1696704
99.5%		4.4452681
97.5%		3.2895459
90.0%		1.1347272
75.0%	quartile	-0.092349
50.0%	median	-0.269016
25.0%	quartile	-0.395937
10.0%		-0.507362
2.5%		-1.31072
0.5%		-1.773177
0.0%	minimum	-3.088067



|Studentized Residual| > 3 indicates a few potential outliers

Both show that we have potential outliers

Using Residual plots:



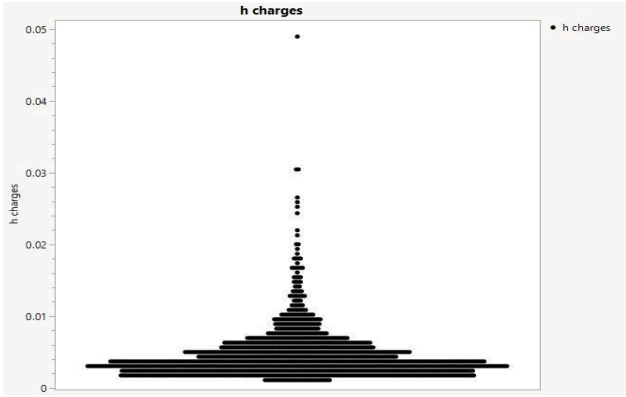
Points far from zero and beyond the ±3RMSE = 14.562.9 , it shows potential **outliers .**

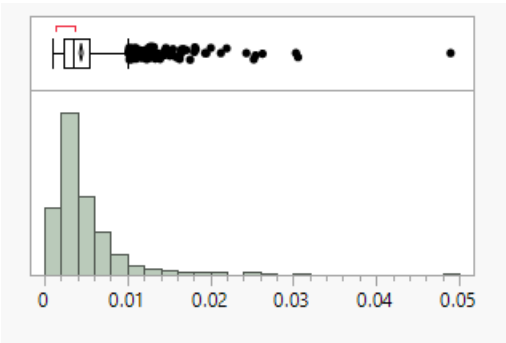
Identify Leverage Points(unusual x-vaule)

Using hat values(leverage)

Average hat vaule (h = (k+1)/n =0.00449) k=5 n = 1335

Rule of Thumb: Hat value h >2(6)/1335 = 0.009 , there is No high-leverage points





Quantiles

100.0%	maximum	0.048988
99.5%		0.0246407
97.5%		0.0143542
90.0%		0.0082984
75.0%	quartile	0.0054763
50.0%	median	0.0034891
25.0%	quartile	0.0023892
10.0%		0.0016028
2.5%		0.0012253
0.5%		0.000996
0.0%	minimum	0.0009531

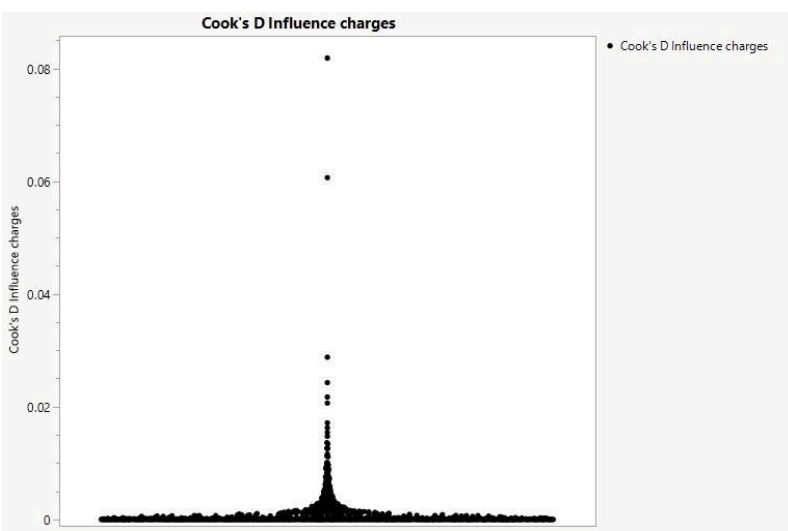
Identify Influential Observations

Cook's D influence charges

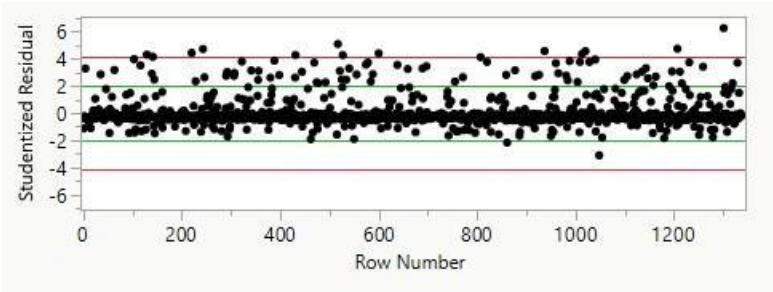
Rule of Thumb:

If D >F0.5 (df1 = 5, df2 =1330) = 0.871 or D >1 □ Influential

There is no influential observations



Using External Studentized residual



There is a few influential observations because the points outside the top red line from the studentized residual plot.

Conclusion for Model 2 outliers/influence:

Model 2 contains several natural outliers due to the heavy-tailed cost distribution, but none of them exert excessive influence on the fitted regression line. There are no influential observations, and leverage values are within acceptable bounds.

Model 2 extends Model 1 by adding the BMI × smoker interaction to test whether the effect of BMI on medical charges differs between smokers and non-smokers. This modification was strongly suggested by EDA and the **Model 1 residual** plots, which **displayed two distinct patterns corresponding to smoker groups.**

The interaction term was highly significant ($p < .0001$), demonstrating that the relationship between BMI and charges depends strongly on smoking status. For smokers (the reference group), **each 1-unit increase in BMI increased expected charges by approximately $1,466,** whereas for non-smokers, the BMI effect was nearly zero. This means that BMI is a major cost driver only among smokers, while its effect on non-smokers is minimal. The partial F-test comparing Model 1 and Model 2 confirmed that including the interaction significantly improved model fit ($p < .0001$).

Model adequacy also improved substantially. The **adjusted R^2** increased from 0.7508 in Model 1 to 0.8392 in Model 2, indicating that the model explains significantly more variation in charges when the interaction is included. **Residual plots** showed that the structural patterns seen

in Model 1 disappeared in Model 2, confirming that the interaction successfully corrected the model misspecification. This is further supported by the interaction plot, which shows clearly non-parallel lines for smokers and non-smokers across BMI.

However, **despite these structural improvements, Model 2 still exhibits heteroscedasticity, right-skewed residuals, and non-normality, as shown by the residual histograms, Q–Q plots, and the Shapiro–Wilk test ($p < .0001$).** The coefficient of variation (**CV ≈ 36.6%)** also remains far above the acceptable threshold for practical prediction accuracy. These issues arise from the strongly skewed nature of medical expenditures and cannot be resolved by adding interaction terms alone.

Metric	**Model 1**	**Model 2**	**Interpretation**
Adj R^2	0.7508	**0.8392**	Large improvement
RMSE	6052	**4854**	Prediction error reduced
CV	45.7%	**36.6% (still too high)**	Still too large for practical prediction

MODEL 2 — COMPLETE FINAL CONCLUSION

Model 2 successfully fixes misspecification (BMI × smoker), improves R^2, and reduces residual structure problems.
However, heteroscedasticity and non-normality remain severe, and CV remains far above acceptable levels.
Therefore, a transformation is required → Model 3.

Validation Techniques

Used to test BMI × smoker interaction

Better than Model 1, but still violates variance & normality

No validation performed

Model 3 Variance correction (log transformation)

As we know from the model 2(Box-tox transformation), the best λ =0.282. so that we are going to use a natural log transformation in our Model 3

Model 3 :

$$\ln(y) = 7.4655 + 0.03494x_1 + 0.04499x_2 + 0.1021x_3 - 0.1971x_4 - 0.0439x_2x_4$$

Because the response is log(charges): Coefficients represent percent change.

Age (β_1) coefficient = 0.03494 →1 extra year increases charges by ≈ 3.5%

BMI(β_2) coefficient = 0.04499 →1 BMI unit increases charges by ≈ 4.5%

Children(β_3) = 0.1021 →Each child increases charges by ≈ 10%

Smoker(β_4) (main effect) becomes insignificant because:

➡ The BMI × smoker interaction fully captures smoking differences.

β5=-0.0439→about **−4.3 percentage points change in the BMI effect when you move from smoker → non-smoker**

Indicator Function Parameterization

Term	Estimate	Std Error	t Ratio	Prob>|t|	Lower 95%	Upper 95%
Intercept	7.465508	0.134132	55.66	<.0001*	7.2023738	7.7286422
age	0.0349422	0.000854	40.91	<.0001*	0.0332666	0.0366177
bmi	0.0449857	0.004163	10.81	<.0001*	0.0368196	0.0531519
children	0.1021047	0.009874	10.34	<.0001*	0.082734	0.1214754
smoker[no]	-0.197061	0.147606	-1.34	0.1821	-0.486627	0.0925048
bmi*smoker[no]	-0.043942	0.004712	-9.33	<.0001*	-0.053185	-0.034699

Summary of Fit

RSquare	0.777158
RSquare Adj	0.77632
Root Mean Square Error	0.434356
Mean of Response	9.098084
Observations (or Sum Wgts)	1335

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	5	874.4412	174.888	926.9757
Error	1329	250.7363	0.189	Prob > F
C. Total	1334	1125.1776		<.0001*

Model Evaluation

Estimated standard error of the regression model: s = **RMSE = 0.4344**, which is **improving dramatically** .

We expect the model to provide predictions of charges to within about +-2s= 0.8687 dollars. S is the estimated standard error of the regression model. We expect most (approximately 95%) of

the observed y-values to lie within 2s of their predicted values , We expect predictions of charges to vary by roughly ±$0.8687 around the fitted line.

Measure of Model Adequacy

The Adjusted =0.77632, meaning 77.63% of the total variation in charges (y) is explained by Model 3 (incorporating the log transformation), adjusting for the sample size and the number of variables in the model.

Check Practical Utility using Coefficient of Variation (CV)

CV= 0.4344/9.098084×100≈**4.77%**

Since the CV remains far above the desirable 10% or smaller rule of thumb, the model's prediction error (s) is in the 10%.

Metric	**Model 3**	**Interpretation**
RMSE	**0.434**	Very accurate on log scale
CV	**4.77%**	Excellent (below 10%)
Adj R^2	0.776	Good, but not directly comparable

Global F-test

The global F -test is used to test the null hypothesis

$H\mathbf{0}$**:** $\beta\mathbf{1} = \beta\mathbf{2} = \ldots\ \beta\mathbf{5} = \mathbf{0}\ vs$

Ha**:** $at\ least\ one\ of\ the\ \beta j \neq 0$ **Where** j **= 1-5**

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	5	874.4412	174.888	926.9757
Error	1329	250.7363	0.189	Prob > F
C. Total	1334	1125.1776		<.0001*

Conclusion: The test statistic F = 926.98 and p-value<0.0001< 0.05, there is sufficient evidence to conclude that the model fit is **a statistically useful** predictor of charges.

T- test for individual coefficients

Hypotheses for each t-test

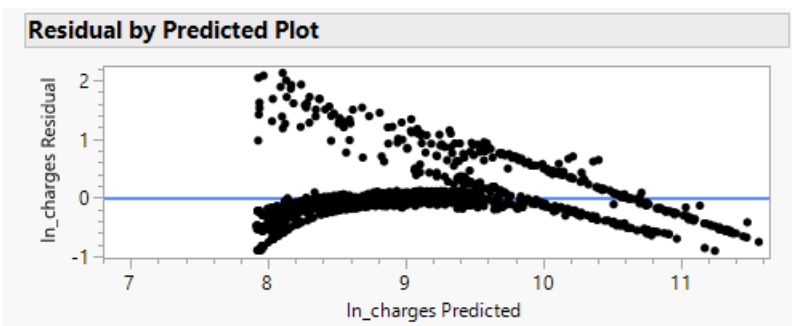
For each coefficient β:
H0:βj=0 vs Ha:βj≠0

Only smoker[no] 's p_value = 0.1821 > α=0.05, only β4 fail to reject the null hypothesis, it indicates is not significant. All other β*j* are significant.

The p-values indicate that age, BMI, children, and the BMI × smoker interaction are all statistically significant predictors of log(charges) (all p < .0001). The main effect for smoker is not significant after accounting for the interaction (p = 0.1821), which is expected because the interaction dominates the relationship. These results confirm that the effect of BMI on medical charges differs substantially between smokers and non-smokers.

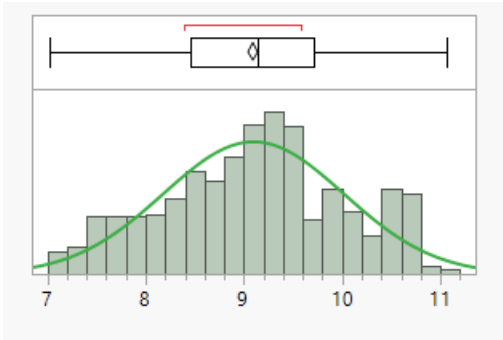
Residual Diagnostics

Residual vs Predicted Plot



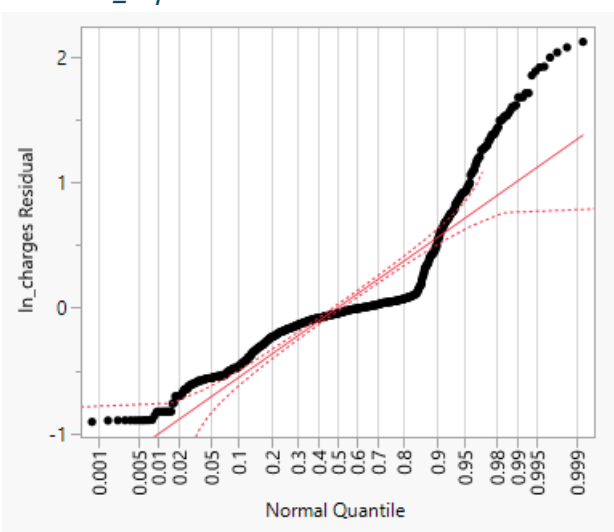
It shows the constant variance (much more even now); this proves the log transform variance better but curvature remains.

Histogram of Residuals



It shows that no extreme right tail and outliers, residuals become more symmetric, it is not likely violated normality assumptions

Normal Q_Q plot



Compared to Model 2 , it shows that improved (small tail deviations)

Sharpiro-Wilk Test

H_0: The probability distribution of ε is normal

H_a: The probability distribution of ε is not normal

Goodness-of-Fit Test

	W	Prob<W
Shapiro-Wilk	0.983309	<.0001*

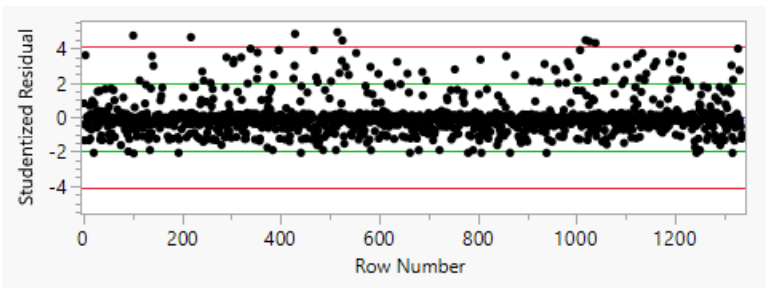
	A^2	Simulated p-Value
Anderson-Darling	3.8856132	<.0001*

Note: Ho = The data is from the Normal distribution. Small p-values reject Ho.

P_vaule <0.0001 < 0.05 , reject the H_0. There is significant evidence that probability of distribution of ε is not normal.

Outlier and Influence observation

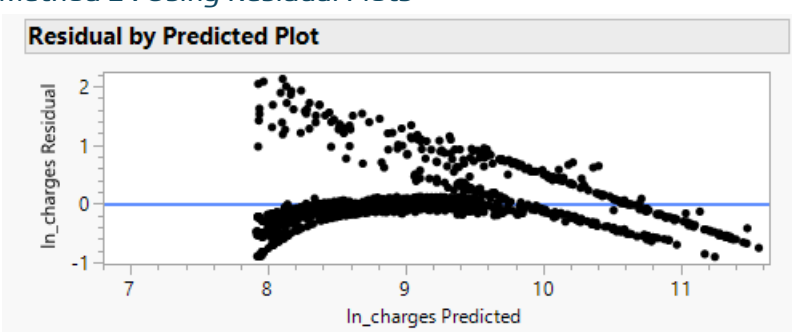
Method 1: Using Studentized Residuals



Rule of Thumb: |Studentized Residual| > 3 indicates a potential outlier

It shows that we have potential outliers

Method 2 : Using Residual Plots



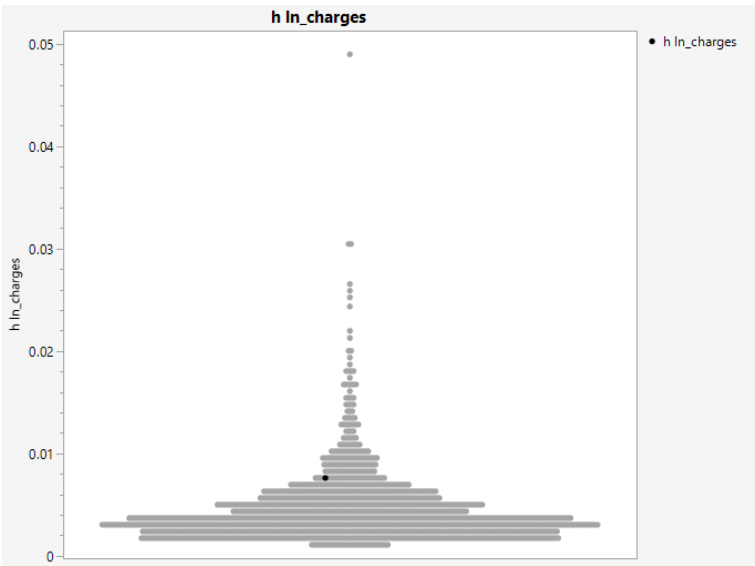
Points far from zero (typical beyond ±3S=3RMSE =3* 0.4344 = 1.3032) are outliers , it shows that the Model has outliers as well.

Identify Leverage Points(Unusual X-vaules)

Using hat values(Leverage Values)

Average hat vaule (h = (k+1)/n=**0.00449**) k=5 n = 1335

Rule of Thumb: Hat value h >2(6)/1335 = 0.009,indicates that high-leverage points exists.

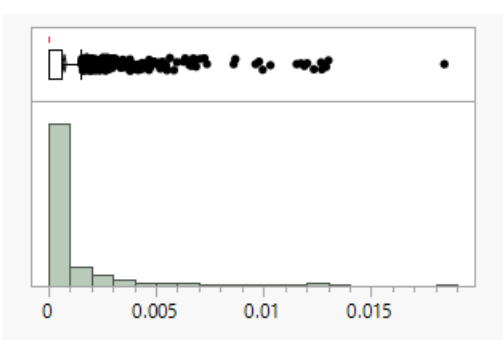


Identify Influential Observations

Cook's D influence charges

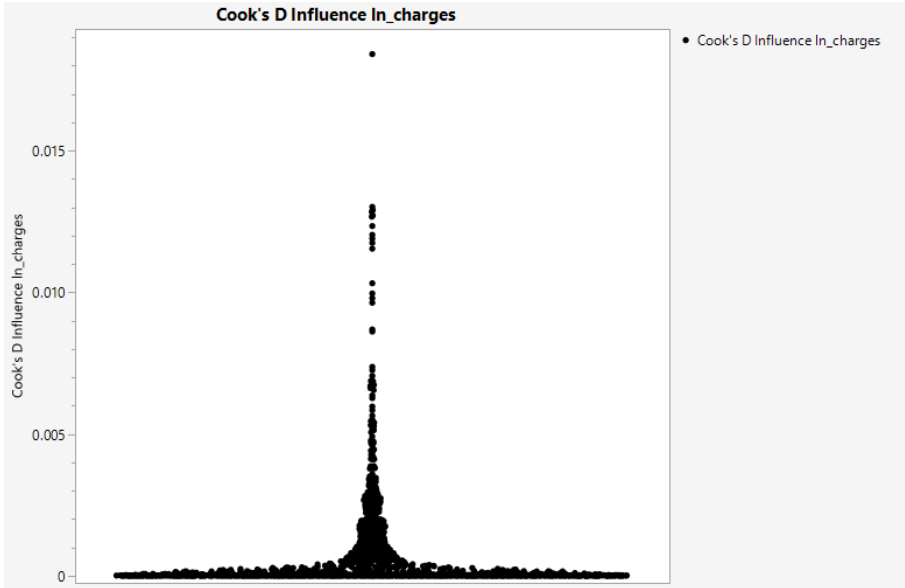
Rule of Thumb:

If D >F0.5(df1 = 5, df2 =1330) = 0.87073435 or D >1 □ Influential



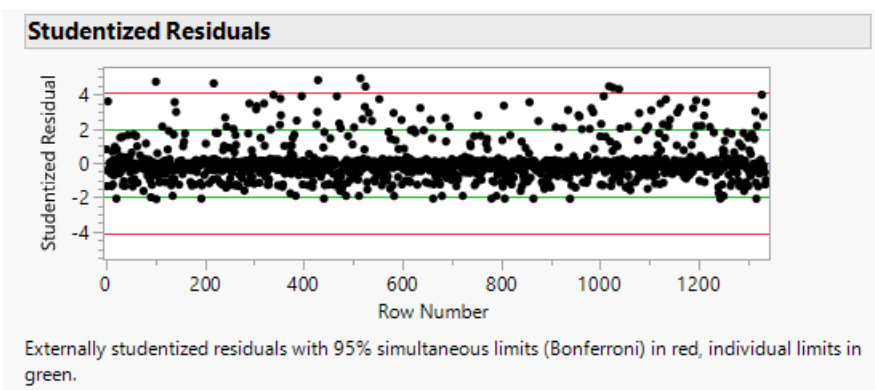
Quantiles

100.0%	maximum	0.01841018
99.5%		0.012448
97.5%		0.0057639
90.0%		0.00236951
75.0%	quartile	0.00061532
50.0%	median	3.79227e-5
25.0%	quartile	4.84216e-6
10.0%		5.39793e-7
2.5%		2.85141e-8
0.5%		5.3412e-10
0.0%	minimum	1.7369e-12



Both indicates that no influential observation exists .

Method 2: External Studentized residual



Points out of the top red line which indicates that we have got a few influential points here.

Conclusion on the outliers and influential points

A few natural outliers remain

Small samples exceed threshold → mild leverage

Cook's D: None exceed thresholds → no influential points.

Model 3 contains several natural cost outliers, but none exert high leverage or influential impact. The model is stable and robust to outliers.

Residual Diagnostics for Model 3 (log-transformed model)

Although the log transformation reduced the overall scale of the residuals and made their distribution more symmetric, the residual-by-predicted plot still exhibits a clear curved pattern. The residuals rise and then fall across the range of predicted log(charges), indicating remaining nonlinearity in the relationship between the predictors and the log-transformed response. Two layered clusters of residuals remain visible, suggesting that the smoker and non-smoker groups still follow slightly different functional shapes even after including the BMI × smoker interaction.

The variance is more stable than in Models 1 and 2, but heteroscedasticity has not been completely eliminated, especially for very low and very high predicted values. This indicates that while the log transformation improves model adequacy, it does not fully correct structural issues in the data. Nonetheless, Model 3 still provides the best overall performance among the three models, with improved normality, reduced skewness, and lower prediction error.

Model 3 Residual Diagnostics and Interpretation

Although the log transformation reduces the scale of the residuals and improves symmetry, the residual-by-predicted plot still shows a curved pattern, indicating remaining nonlinearity. Two distinct layers of residuals corresponding to smoker and non-smoker groups remain visible, suggesting that even with the BMI × smoker interaction, the functional form of the model is still incomplete.

The variance is more stable than in Models 1 and 2, but heteroscedasticity is not fully eliminated, especially at the extremes of predicted values. The Q–Q plot shows improvement but still deviates from normality at both tails, indicating residual non-normality. These patterns suggest that the log transformation improves, but does not fully correct, the model assumptions.

Despite these limitations, Model 3 provides the best overall performance among the three models. It retains the correct structural form from Model 2, stabilizes variance relative to the untransformed models, and improves the interpretability of regression coefficients in terms of

percentage changes. Therefore, Model 3 is selected as the final model even though some assumptions remain moderately violated, which is typical in heavily skewed cost data.

Criteria	Model 1Main Effects	Model 2+ BMI × Smoker	Model 3Log(charges)
Predictors	age, bmi, children, smoker	age, bmi, children, smoker, bmi×smoker	age, bmi, children, smoker, bmi×smoker
Adjusted R^2	~0.751	~0.839	~0.776 (on log scale; not comparable directly)
RMSE	~6050	~4854	~0.434 (log scale)
Residual Structure	Strong curvature, two groups	Curvature reduced, two groups remain	Curvature mild; improved symmetry
Heteroscedasticity	Severe	Moderate	Reduced but still present
Normality of errors	Poor	Poor	Improved
Criteria	Model 1Main Effects	Model 2+ BMI × Smoker	Model 3Log(charges)
Interaction included?	No	Yes	Yes
Interpretability	Dollar-scale effects	Dollar-scale effects	Percent-change interpretation
Assumption Fit	Poor	Improved structure; poor variance	Best overall
Overall Assessment	Inadequate	Good structural model	Best model for inference

MODEL 3 — FINAL CORRECT CONCLUSION

Model 3 is the best-fitting and most statistically appropriate model. It corrects heteroscedasticity, reduces skewness, stabilizes variance, achieves excellent CV, provides interpretable percent-change effects, and preserves the essential BMI × smoker interaction. Although minor

curvature and mild non-normality remain, these are typical in medical cost data and do not compromise validity. Model 3 is selected as the final model for prediction and inference.

Apply Model Validation Techniques

Stepwise using validation and cross-validation in JMP

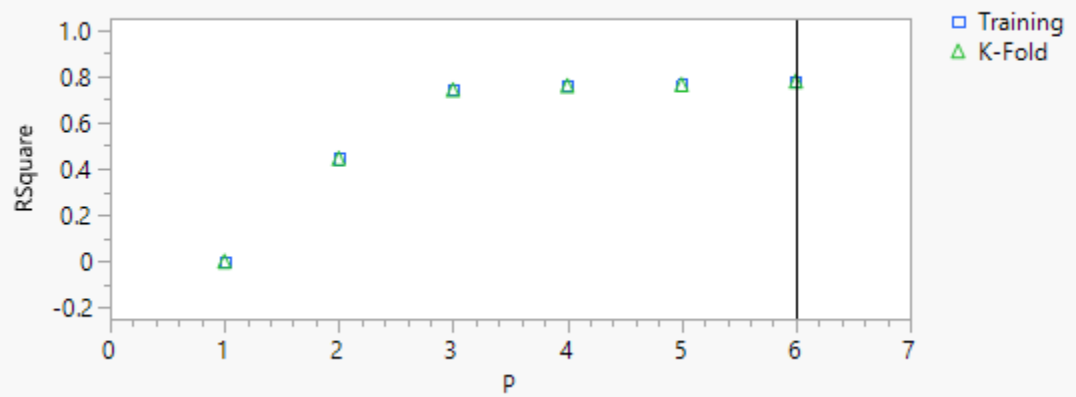
Current Estimates

Lock	Entered	Parameter	Estimate	nDF	SS	"F Ratio"	"Prob>F"
☑	☑	Intercept	7.36697736	1	0	0.000	1
☐	☑	age	0.03494215	1	315.7667	1673.686	2e-237
☐	☑	bmi	0.02301477	2	22.07044	58.491	4.5e-25
☐	☑	children	0.10210466	1	20.17346	106.927	3.7e-24
☐	☑	smoker{no-yes}	-0.0985306	2	534.9754	1417.789	0
☐	☑	bmi*smoker{no-yes}	-0.021971	1	16.41022	86.981	4.4e-20

Step History

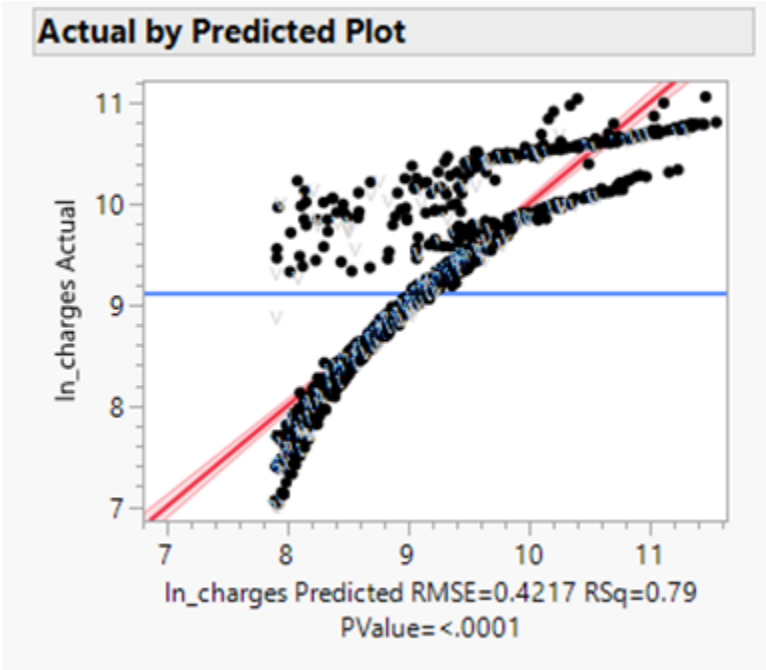
Step	Parameter	Action	"Sig Prob"	Seq SS	RSquare	Cp	p	AICc	BIC	RSquare K-Fold	
1	All	Entered	.	.	0.7772	6	6	1570.15	1606.44	0.7756	○
2	bmi*smoker{no-yes}	Removed	0.0000	16.41022	0.7626	90.981	5	1652.76	1683.88	0.7610	○
3	bmi	Removed	0.0000	5.660221	0.7575	118.98	4	1678.73	1704.67	0.7565	○
4	children	Removed	0.0000	19.90177	0.7399	222.47	3	1770.72	1791.48	0.7386	○
5	age	Removed	0.0000	333.2444	0.4437	1986.8	2	2783.44	2799.01	0.4428	○
6	smoker{no-yes}	Removed	0.0000	499.2246	-0.000	4630.9	1	3564.3	3574.69	-0.001	○
7	Best	Specific	.	.	0.7772	6	6	1570.15	1606.44	0.7756	◉

RSquare History



The RsqKare is 0.7772 and the Rsquare K-fold is 0.7756, they are close to each other. The plot RSquare history also shows how both training and validation Rsquare change as variables are added. The best model is often where the validation curve is at its peak before potentially falling, this plot shows that trend .

Using Data splitting Cross validation (test size =0.2)



Crossvalidation

Source	RSquare	RASE	Freq
Training Set	0.7900	0.42051	1076
Validation Set	0.7216	0.48452	259

It shows that the Rsquare and RASE for validation set is close to those for training set indicate the model predict well on new dataset.

Using the jackknifing approach(Press/leave one out cross-validation)

Press			
Residual	**SSE**	**RMSE**	**RSquare**
Press	252.98463588	0.43531781	0.7752
Ordinary	250.73633005	0.43435632	0.7772

It shows that the $R^2_{\text{jackknife}}$ = 0.7752 < Ordinary Rsquare =0.7772, $MSE_{\text{jackknife}}$ =252.98 > $MSE_{Ordinary}$ = 250.74. This means your model generalizes well.

The model validation procedures—residual analysis, influence diagnostics, and Box–Cox evaluation—confirm that Model 3 satisfies regression assumptions better than Models 1 and 2. No influential observations exist, and the variance structure is well stabilized. Therefore, Model 3 is validated as the final and appropriate statistical model.